

Lecture (21): Hearing Mechanisms

1. Sound

➤ Definition:

Waves of compression & rarefaction

➤ Velocity (V):

V in gases < liquids < solids

N.B. - V in air = 340 m/s at 20°C (↑↑ by temp. & not affected by atmospheric pr.)

- V in endolymph (fluid media of inner ear) = 1360 m/s at body temp.

➤ Characters:

1. Frequency (pitch):

*= N^o of waves / sec

*Depends on wavelength, i.e., distance between 2 successive peaks of sound waves

*Measured in Hertz (Hz)

*Range of human hearing: 20 – 20000 Hz

*Range of average speech voice: 2000 – 5000 Hz

N.B. Males have low pitched voices, while females have high pitched voices

2. Amplitude (intensity):

*Measured in decibel (dB) = 1/10 Bell

*Intensity of sound (dB) = $10 \log \frac{\text{intensity of sound}}{\text{intensity of standard sound}}$

N.B. Standard sound is the sound just heard by normal human ear at frequency 1000 – 3000 Hz (Its intensity = $\log 1 = 0$ dB)

3. Quality (timbre): Depends on waveform, i.e., overtones

2. Human Ear

(A) External ear (EE):

➤ <u>Structure</u>	➤ <u>Function</u>
1. <u>Auricle (ear pinna):</u>	1. Collection of sound waves & directing them towards EAM
	2. Localization of sound “Rotation reflex”
2. <u>External auditory meatus (EAM):</u>	1. Amplification & conduction of sound waves towards middle ear 2. Protection: (a) It secretes wax → lubricates drum & prevents entry of insects (b) It contains hair → traps dust & prevents its entry
3. <u>Tympanic membrane (ear drum):</u> *Shape: - Cone-shaped, with its concave surface directed outwards & downwards - 1 cm in diameter *Characters: Elastic – Tense – Resonator – Aperiodic – Damped	Transmission of sound waves to middle ear

(B) Middle ear (ME):

➤ **Structure:**

Air-filled cavity in the temporal bone, with:

1. 3 Ossicles: From out inwards “Malleus → Incus → Stapes”

2. 3 Muscles:

*2 inside: – Tensor tympani: inserted in handle of malleus, supplied by trigeminal nerve

Stapedius ms.: inserted in neck of stapes, supplied by facial nerve*1 outside:
– Tensor palati: inserted in Eustachian tube (ET), supplied by trigeminal nerve

3. 3 Openings:

*Oval window: closed by footplate of stapes

*Round window: covered by 2ry tympanic membrane (a flexible membrane connected to scala tympani of IE)

*Opening of ET: usually closed – open during swallowing, chewing, yawning & ascent to high altitude

➤ Functions:

1. Transmission of sound waves from EE (vibrating ear drum) to IE *via* oval window

2. Impedance matching: Magnification of intensity of sound waves in ME by ≈ 22 times to overcome the inertia of perilymph in IE:

(a) 1.3 times; due to arrangement of bony ossicles in a lever-like action

(b) 17 times; due to large surface area ratio between tympanic membrane & oval window

3. Contraction of tensor tympani pulls the handle of malleus inwards → ear drum is kept continuously tense → optimum vibration

4. Attenuation (acoustic/tympanic) reflex: Loud sound → contraction of stapedius ms. → prevent inward displacement of stapes inside oval window → protect the IE structures from damage

N.B. It has a long conduction delay (40 – 160 msec) → ineffective if the high-pitched sound is of an extreme onset, *e.g.*, gunshots or jet planes

5. ET → Equalization of pressure on both sides of the ear drum allowing its free movement

N.B. - During descent, the drum may rupture (prevented by chewing or swallowing)

- ET in children is short, wide & straight → Easy infection (OM)

6. 2ry tympanic membrane moves with movements of basilar membrane to

absorb waves created in perilymph of scala vestibuli 2ry to movements of basilar membrane

N.B. Loss of elasticity of 2ry tympanic membrane → hearing loss

(C) Inner ear (IE) = Labyrinth:

➤ **Structure:** (Bony & membranous labyrinth)

1. Auditory labyrinth = Cochlea:

- Responsible for hearing
- Supplied by cochlear division of 8th CN

2. Non-auditory labyrinth = Vestibular apparatus:

- Responsible for equilibrium
- Supplied by vestibular division of 8th CN

Cochlea

35 mm long, 2 $\frac{3}{4}$ turns

Vestibular (Reissner's) membrane & basilar membrane divide it into 3 scala (chambers);

*The upper scala (scala vestibuli) & the lower scala (scala tympani):

- Contain perilymph
- Communicate with each other at the apex of cochlea by a small opening "helicotrema"
- End at the base of cochlea by the oval window & the round window, respectively

*The middle scala (scala media):

- Contains endolymph
- Is continuous with the membranous labyrinth
- Does not communicate with the other 2 scala

Organ of Corti

- Located on the basilar membrane → project into s. media
- Contains "hair cells" which are the auditory mechanoreceptors, arranged in 2 groups:

	Inner hair cells	Outer hair cells
<u>Site</u>	Medial to tunnel of Corti	Lateral to tunnel of Corti

<u>N° of rows</u>	1	3
<u>N° of cells</u>	3500	20.000
<u>Afferent</u>	90 – 95 % of cochlear nerve fibers	5 – 10 % of cochlear nerve fibers
<u>Efferent</u>	Few fibers from sup. olive	Many fibers from sup. olive
<u>Function</u>	The auditory receptors	Improve hearing by affecting vibration of basilar membrane

3. Mechanism of Hearing

(1) Sound transmission:

Sound waves are transmitted by the ear drum & auditory ossicles into the fluid of IE
Sound conduction occurs through:

Ossicular conduction	Air conduction	Bone conduction
Through tympanic membrane & auditory ossicles	Sound waves initiate vibrations in the 2ry tympanic membrane that closes the round window	Through transmission of vibrations of skull bones to fluid of IE
It is <i>the main pathway</i> for normal hearing	It occurs in cases of damage of tympanic membrane & auditory ossicles	It occurs on testing of hearing by applying vibrating tuning fork on the skull

It is <i>efficient</i> due impedance matching	It is <i>inefficient</i>	It is <i>inefficient</i>
---	--------------------------	--------------------------

(2) Stimulation of hair cells:

Movement of stapes → movement of oval window (in & out) → series of waves in the perilymph of scala vestibuli & scala tympani → vibrations of the basilar membrane (up & down) → movement of the organ of Corti (up & down):

(a) If it moves up:

Stereocilia bend away from the limbus → K^+ channels open → K^+ inflow → depolarization of hair cells → Ca^{2+} channels open → Ca^{2+} inflow → release of chemical transmitter (glutamate or aspartate) → stimulation of auditory nerve fibers

(b) If it moves down:

Stereocilia bend towards the limbus → K^+ channels close → no K^+ inflow → hyperpolarization of hair cells → the reverse

(3) Transmission of AP in auditory nerve fibers: “Auditory pathway”

Each cochlea is ***bilaterally represented*** in both temporal lobes

*Receptors: Inner hair cells of the organ of Corti

*1st order neuron: the cells of spiral ganglion

Axons: form the cochlear division of 8th CN → enters the brainstem & ends in medulla

*2nd order neuron: Dorsal & ventral cochlear nuclei (in the medulla)

Axons: transmit auditory impulses through different pathways & may end on or bypass the following nuclei:

1. Sup. olivary nucleus & nucleus of trapezoid body ***on both sides***
2. Nucleus of lat. lemniscus

3. Inf. Colliculus in the midbrain

*3rd order neuron: Med. geniculate body (in the thalamus)

Axons: form the auditory radiations to the center

*Center: the auditory cortex:

- **1^{ry} auditory cortex**: areas 41 & 42 (in the sup. temporal gyrus)
 - Functions:
 1. Recognition of sound pitch, intensity & quality
 2. Sound localization
 - Lesion:
 1. Unilateral lesion → only slight ↓ of hearing in the opposite ear (Why?)
 2. Bilateral lesion → severe ↓↓ of hearing with abolishment of sound pitch & localization
- **Auditory association area**: area 22
 - Function: Understanding (interpretation) of the meaning of heard sounds
 - Lesion: No interpretation of the meaning of heard sounds

(4) Auditory encoding:

*Definition: It is the ability of auditory system to discriminate sound frequency, intensity, locality & patterns

1. Frequency (pitch) discrimination:

(a) Place (resonance) theory:

- Discrimination of sound frequency is a function of cochlea and not the auditory cortex which only interprets various frequencies as different pitches.
- The basilar membrane is a V-shaped fibrous membrane containing about 20000 elastic fibers embedded in a gelatinous material. It extends from the base to the apex of the cochlea in a spiral form. The organ of Corti is located on it, and is

stimulated by its vibration.

- The theory postulates that ‘the fibers of the basilar membrane act as resonators; each group of these fibers can be set into vibration only in response to a particular sound frequency’

- The major determinant of sound pitch is the **‘place’** of the organ of Corti at the basilar membrane that is maximally stimulated by sound waves.

- High frequency sounds produce resonance in the short stiff fibers at the base of the cochlea, while low frequency sounds produce resonance in the long lax fibers at its apex.

(b) Travelling wave theory: **(the most accepted theory)**

- It is a modification of the place theory since it also considers the cochlea to be the site of pitch analysis.

- It states that ‘the travelling wave in the perilymph of scala vestibuli causes maximum deflection of the basilar membrane at a specific distance from the oval window that is characteristic for each given frequency’.

- The distance is inversely related to the sound pitch:

- i. Low frequencies → maximal stimulation at the apex of the cochlea

- ii. Higher frequencies → maximal stimulation at the base of the cochlea

(c) Tonotopic organization:

Nerve fibers & nuclei of the auditory pathway show spatial organization all the way from the cochlea to the auditory cortex, e.g., neurons responding to low frequency are located on one side of nuclei & tracts, while those responding to high frequency are located on the other side

2. Intensity (loudness) discrimination:

- It is a function of auditory cortex

- High loud sounds cause:

- a. ↑↑ frequency of firing (A.P.) in auditory nerve fibers
- b. Stimulation of larger portion of the basilar membrane

3. Locality discrimination:

- a. Intensity difference: The sound is louder on the ear closer to its source
- b. Phase difference: At low frequencies, there is phase (time) difference in the sound waves reaching the 2 ears
- c. The ear pinna helps in discrimination of the sound source

4. Pattern discrimination:

Auditory cortex can recognize a sequence of tones following each other in a particular pattern, *e.g.*, music

Lecture (22): Chemical Senses (Smell & Taste)

(Dr. Wagdy Nashaat Habeeb)

- Generally classified as visceral senses due to their close association with GI functions
- Physiologically, they are related to each other:
 - (1) Flavors of foods are a combination of their taste & smell
 - (2) Both of taste & smell receptors are chemoreceptors which are stimulated by molecules
- Anatomically, they are quite different:
 - (1) Smell receptors are distant receptors (teleceptors), its pathway has no relay